

Tree Ring Climate Detective

Practice reading stylized rings as proxy evidence

At a glance: Use an authored ring-card code to build and defend a model history, then explain what real evidence would require. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain that earlywood and latewood form one annual ring and that width-climate relationships depend on species, site, and calibration (Understand).
- Apply the printed code to the six stylized authored cards and cite the exact annual bands that support each model event (Apply).
- Analyze which model history best fits the authored pattern and distinguish that exercise from real climate reconstruction (Analyze).
- Defend a practice-card claim and name the cross-dating, replication, and local calibration real dendrochronology requires (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- printed ring cards: one set per team (4, plus 1 spare), laminated for reuse
- answer boards: one per team (4, plus 1 spare), reusable
- wood cookies (optional): one sanded slice per team if used (4, plus 1 spare); the printed cards are the primary medium
- timers: one per team (4) or the projected room timer
- claim-evidence cards: a stack per team (about 5 to 7 each), printed on the shared cardstock line

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up 4 stations, one per team, each with one ring-card set, one answer board, one timer, a stack of claim-evidence cards, markers, the ring-code legend, and (if used) one sanded wood cookie, sorted and ready.
- 02 Prepare a scoring reference for consistent use of the printed practice-card code and cited bands; do not present the authored cards as cross-dated samples or a real climate answer key. Stage it with the score slips.
- 03 No PPE is needed: this is a dry card-and-board game with no soil, water, or chemicals. If you use real wood cookies, sand them smooth and inspect for splinters and loose bark beforehand, and skip them for any student with a wood-dust sensitivity.
- 04 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Use the authored ring-card code to build a model history, then explain what real evidence would require." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before reading the cards, each team predicts which authored pattern is the clearest model-stress code and explains why. No reading the sequence yet.

Phase 3 Constraints

0:12 - 0:18

Read annual bands from center to bark, apply only the printed practice-card code, and cite exact bands for every model inference. State that alternating fill only separates years and that real interpretation requires cross-dating and local calibration.

Phase 4 Read and infer

0:18 - 0:44

Teams read the authored band sequence from center to bark, record each pattern, and assign model events under the printed code. Circulate and ask which bands support each call without treating the cards as real samples.

Phase 5 Cite the evidence

For each model inference, teams fill a claim-evidence card naming the exact annual bands and pattern that support the call.

Phase 6 Synthesize and re-check

Each team combines its model inferences into one practice-card history, rechecks its least certain claim, and lists the real cross-dating and calibration evidence that is missing.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, then wipe the answer board, return the ring cards and any wood cookies to the bin, and restock blank claim-evidence cards for the next team. No PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team names model events without pointing to bands. **Fix:** Require the exact annual band that supports each claim before moving on.

Problem: A model inference has no cited bands. **Fix:** Use the claim-evidence card to name the exact annual bands and pattern; an unsupported claim does not score.

Problem: Two teams apply the card code differently. **Fix:** Re-read from center to bark and cite exact bands. Explain that real dendrochronologists cross-date many trees and calibrate measurements against local records rather than resolving a real history from one stylized card.

Problem: A team finishes early. **Fix:** Ask them to defend the model history with specific bands and list two kinds of real evidence needed to test it.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to rings as proxy data.

Tier 2 (applying the card code but not reasoning):

- "Which authored annual band or pattern most changes your model history under the card code, and why?"

Tier 3 (ready to extend):

- "Explain how cross-dating many real trees and calibrating ring measurements against local weather could test this card-based model."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Practice-code inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	100

CLEANUP AND SOURCES

Return the ring cards, answer boards, and any wood cookies to the bin, restock blank claim-evidence cards, wipe the dry-erase boards, and collect score slips. There is nothing wet or chemical to dispose of.

SOURCES NOAA NCEI Tree Ring product and paleoclimatology (<https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring>); NOAA Climate.gov, How tree rings tell time and climate history (<https://www.climate.gov/news-features/blogs/beyond-data/how-tree-rings-tell-time-and-climate-history>); UCAR Center for Science Education, Tree Rings and Climate (<https://scied.ucar.edu/learning-zone/how-climate-works/tree-rings-and-climate>).